3D Simulation of Ammonia Gas Dispersion in Chemical Process Industries Using CFD Model

Presented by:
Srajan Mishra (2211701177)

Under the guidance of:
Dr. Bharat K. Modhera
(Department of Chemical Engineering, MANIT Bhopal)

Introduction

- Ammonia is a widely used industrial gas with toxic, flammable, and corrosive nature.
- Industrial accidents such as the Bhopal Gas Tragedy highlight the need for gas dispersion modeling.
- CFD simulations help predict toxic cloud spread under different environmental conditions.
- This study uses ANSYS CFX & SimScale to simulate and analyze ammonia dispersion.

Problem Statement & Motivation

Chemical plants store ammonia under high pressure, risking accidental releases.

Physical experiments are costly, unsafe, and limited in scope.

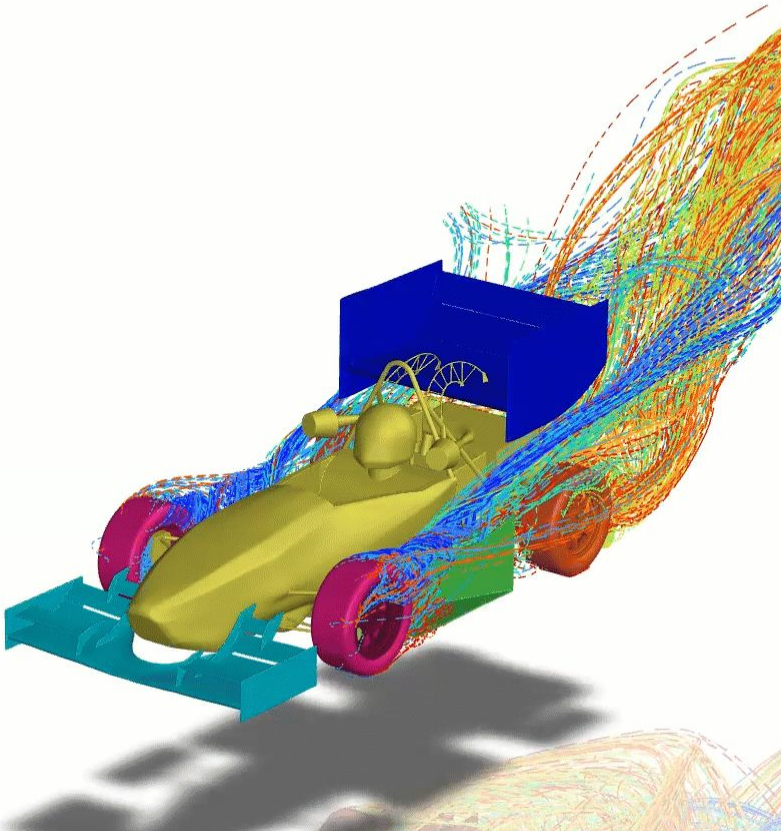
CFD provides a safe, cost-efficient virtual environment to study toxic gas behavior.

There is a need for a validated, physics-based simulation model that can predict ammonia dispersion under realistic plant conditions.

Gas dispersion depends on multiple dynamic factors which cannot be captured through basic analytical models.

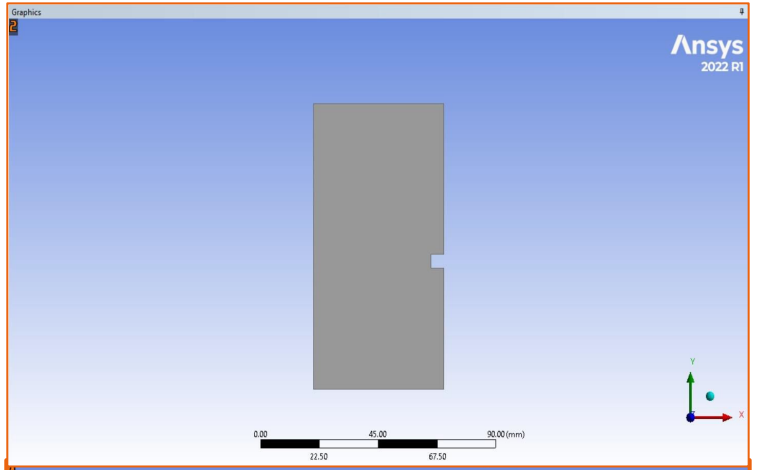
About CFD

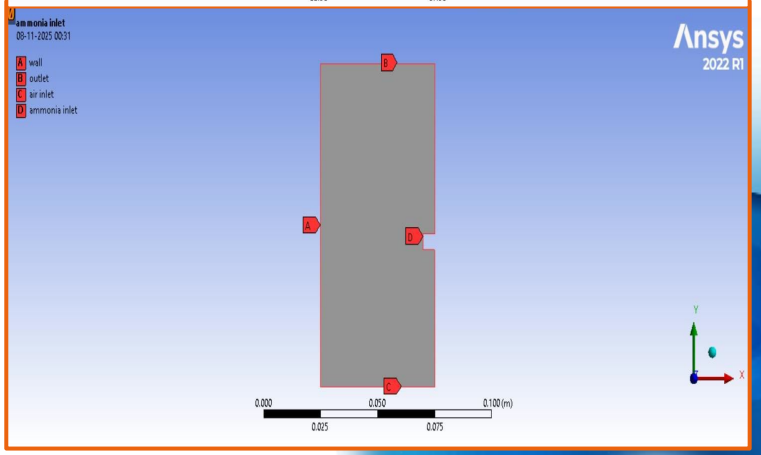
CFD or Computational Fluid Dynamics is a field of simulation based physics which solves Navier–Stokes equations to predict fluid flow behavior.
ANSYS & **SimScale** are software systems which integrates geometry creation, meshing, setup, simulation, and visualization.
The combination allows multi-physics simulations and high-accuracy modeling of gas dispersion.



RECAP

1. Geometry creation for a 50x100 m industrial area.
2. Meshing with coarse, medium, and fine grids.
3. Setup: Ideal gas models, transient simulation (200 s), NH_3 release 5–10 kg/s.
4. Boundary conditions: wind velocity 0.1–4 m/s, temperature 280–300 K.
5. Post-processing: contours, streamlines, pathline animations.





File | Domain | Physics | User-Defined | Solution | Results | View | Parallel | Design

Mesh: Display... | Info | Units... | Check | Quality | Scale... | Transform | Make Polyhedra

Zones: Combine | Separate | Adjacency... | Delete... | Deactivate... | Activate... | Append | Replace Mesh... | Replace Zone...

Interfaces: Mesh... | Overset...

Mesh Models: Dynamic Mesh... | Mixing Planes... | Gap Model...

Turbomachinery: Turbo Models | Turbo Create... | Turbo Workflow | Turbo Topology... | Periodic Instancing...

Adapt: Manual... | Controls... | Automatic... | Manage...

Surface: Create | Manage...

Quick Search (Ctrl+F)

Outline View

Filter Text

- Solution
  - Methods
  - Controls
  - Report Definitions
  - Monitors
  - Cell Registers
  - Automatic Mesh Adaption
  - Initialization
  - Calculation Activities
  - Run Calculation
- Results
  - Surfaces
  - Graphics
    - Mesh
    - Contours
    - Vectors
    - Pathlines
    - Particle Tracks
  - Plots
    - Data Sources
    - XY Plot
      - xy-plot-5
      - xy-plot-4
      - xy-plot-3
      - xy-plot-2
      - xy-plot-1
    - Histogram
    - Profile Data
    - Interpolated Data
    - FFT
    - Cumulative Plot
  - Scene
  - Animations
    - Playback
    - Scene Animation
    - Sweep Surface

Task Page

Run Calculation

Check Case... | Preview Mesh Motion...

Time Advancement

Type: Fixed | Method: User-Specified

Parameters

Number of Time Steps: 1000 | Time Step Size [s]: 1e-6

Max Iterations/Time Step: 19 | Reporting Interval: 1

Profile Update Interval: 1

Options

- Extrapolate Variables
- Report Simulation Status

Solution Processing

Statistics

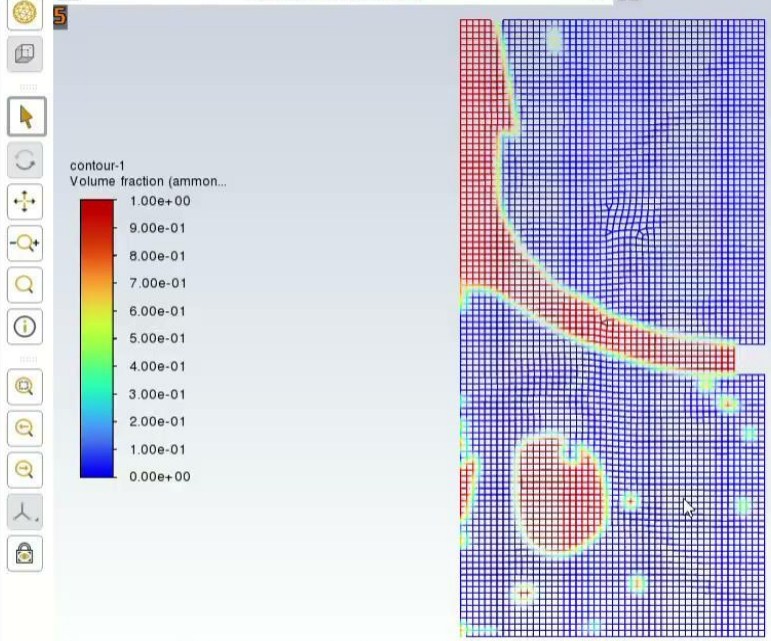
- Data Sampling for Time Statistics

Data File Quantities...

Solution Advancement

Calculate

Volume fraction (ammonia)



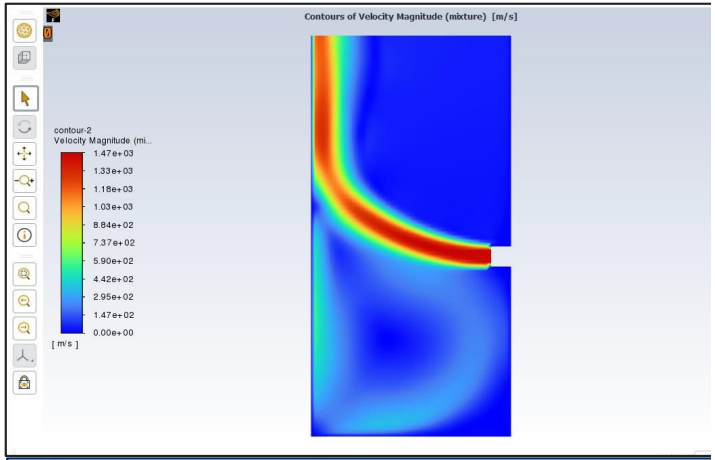
Reserved Window 1

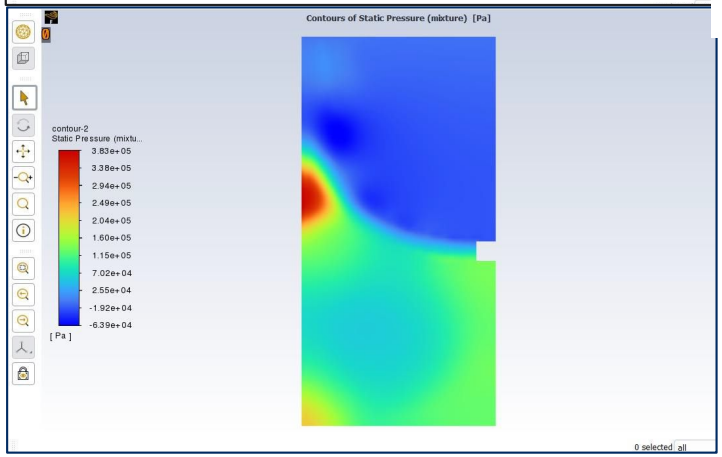
0 selected | all

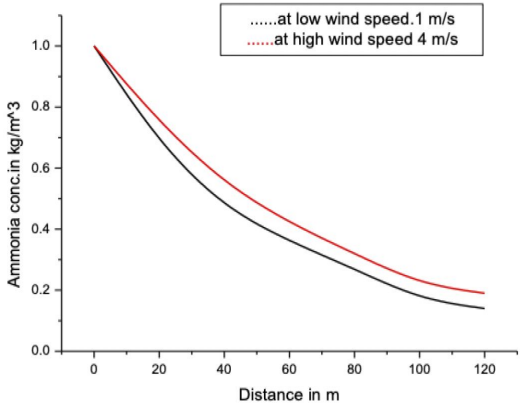
Console

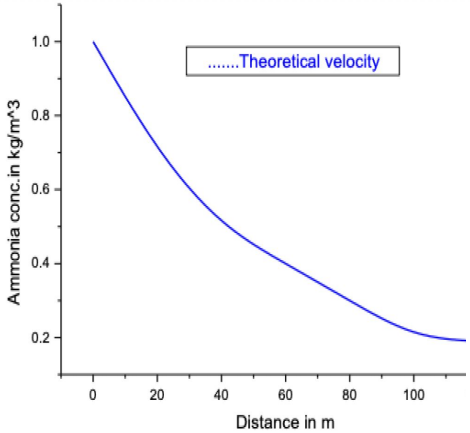
```
Done.

Reading from LAPTOP-M2TVG0EI:"E:\ammoni\ammonia project_files\dp0\FFF\Fluent\FFF.1-10-01000.dat.h5" in NODE0 mode ...


  Reading results.
Parallel variables...
Done.
```

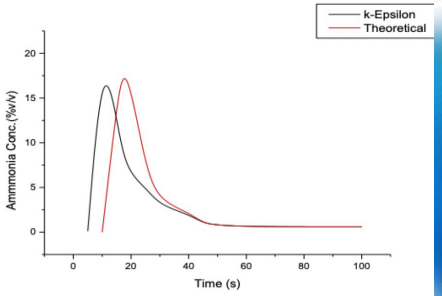
Results

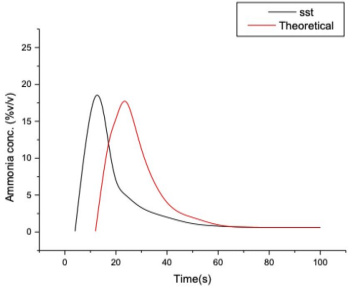

Contours of Velocity Magnitude (mixture) [m/s]
contour-2
Velocity Magnitude (mi...
1.47e+03
1.33e+03
1.18e+03
1.03e+03
8.84e+02
7.37e+02
5.90e+02
4.42e+02
2.95e+02
1.47e+02
0.00e+00
[m/s]

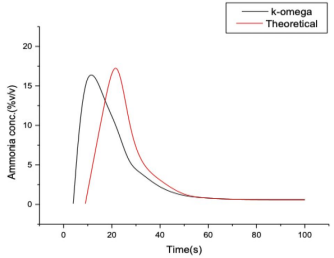

Contours of Static Pressure (mixture) [Pa]
contour-2
Static Pressure (mixtu...
3.83e+05
3.38e+05
2.94e+05
2.49e+05
2.04e+05
1.60e+05
1.15e+05
7.02e+04
2.55e+04
-1.92e+04
-6.39e+04
[Pa]
0 selected all


.....at low wind speed.1 m/s
.....at high wind speed 4 m/s
Ammonia conc.in kg/m^3
Distance in m


.....Theoretical velocity
Ammonia conc.in kg/m^3
Distance in m

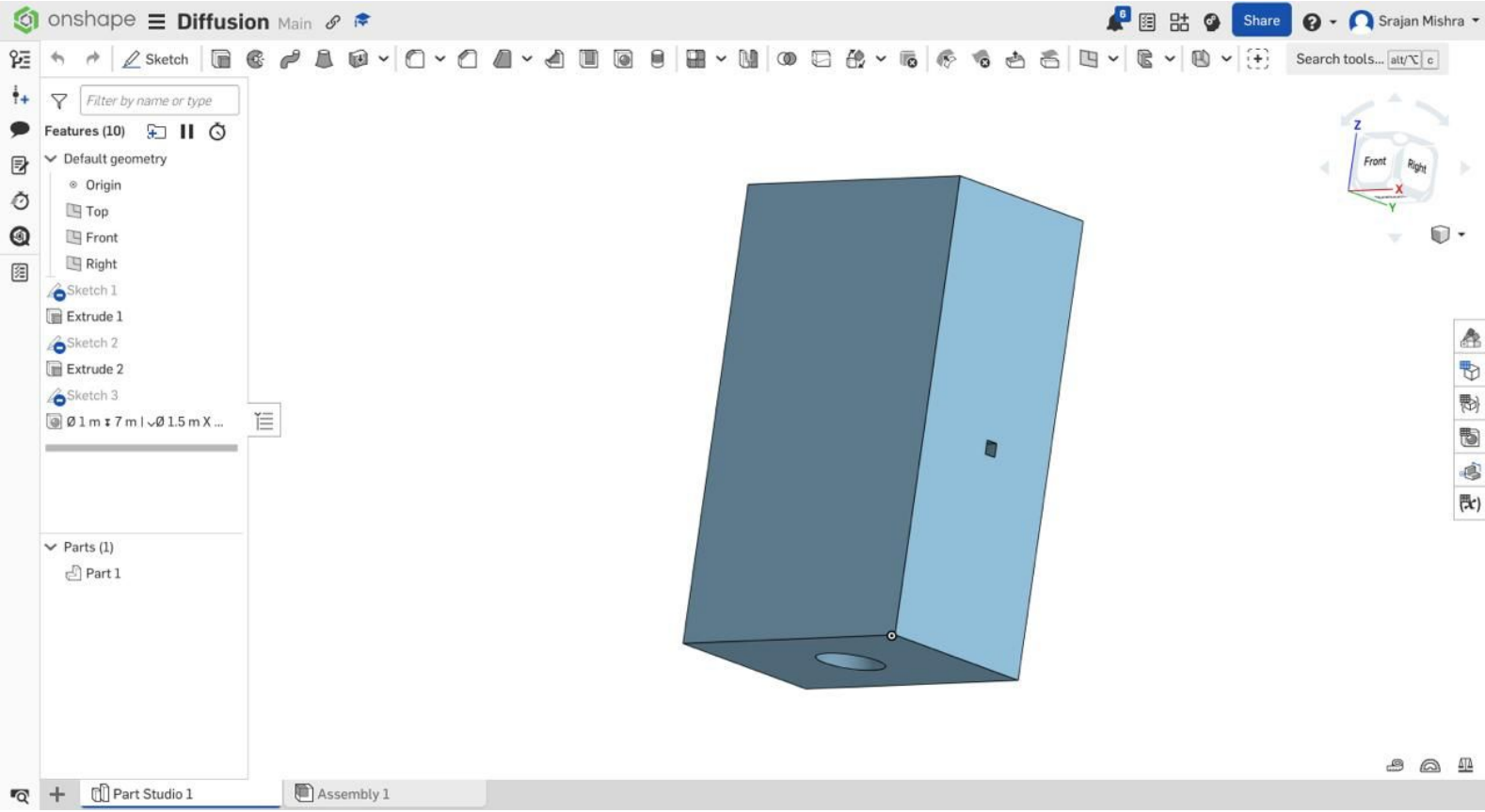

k-Epsilon
Theoretical
Ammmonia Conc.(%v/v)
Time (s)


sst
Theoretical
Ammonia conc. (%v/v)
Time(s)


k-omega
Theoretical
Ammonia conc.(%v/v)
Time(s)

Creating 3D Model

The 3D geometry of the room, obstacle, and leakage opening was created using Onshape CAD software.



Generating Mesh

Table 3.1 The specification and the dimensions are given for model Geometry

Name	Initial X	Initial Y	Initial Z	End X(breadth)	End Y(height)	End Z(length)
Box	0	0	0	500	200	500
Cut Box1	120	0	230	30	30	20
Cut Box2	230	0	250	30	20	35
Cut Box3	190	0	280	20	25	30
Cut Box4	130	0	300	25	20	20
Cut Box5	320	0	150	20	20	30
Cut Cylinder	250	0	110	0	10 Φ=1	0
Cut Tetrahedron1	230	0	210	25	35	30 x 20
Cut Tetrahedron2	200	0	220	30	35	35 x 20
Cut Tetrahedron3	280	0	230	25	30	30 x 20

the cut word specify for the obstacle geometry and box model made of 500 (length) X 500 (breadth) X 200(Height) and in the Figure 3.2 for showing this table geometry in 3D view and the obstacles are named as in table 3.1.

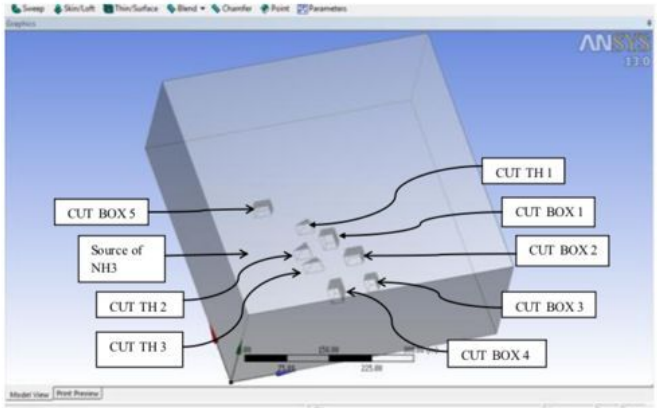


Figure 3. 2 Showing 3D models Geometry in Ansys/CFX

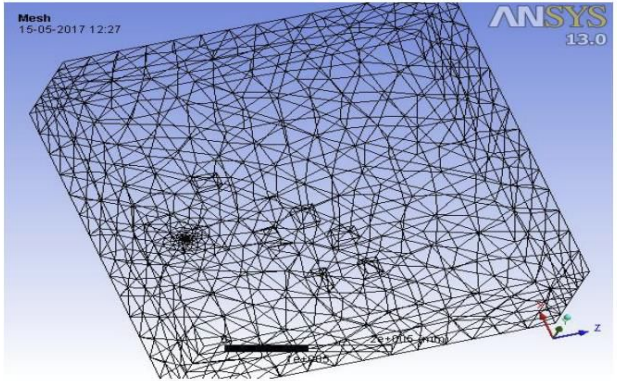


Figure 3. 3 Coarse type of meshing of model Geometry

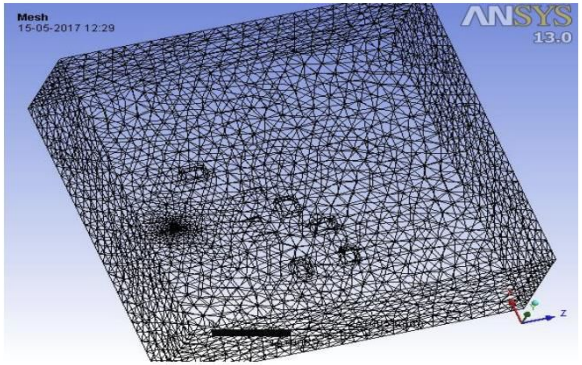
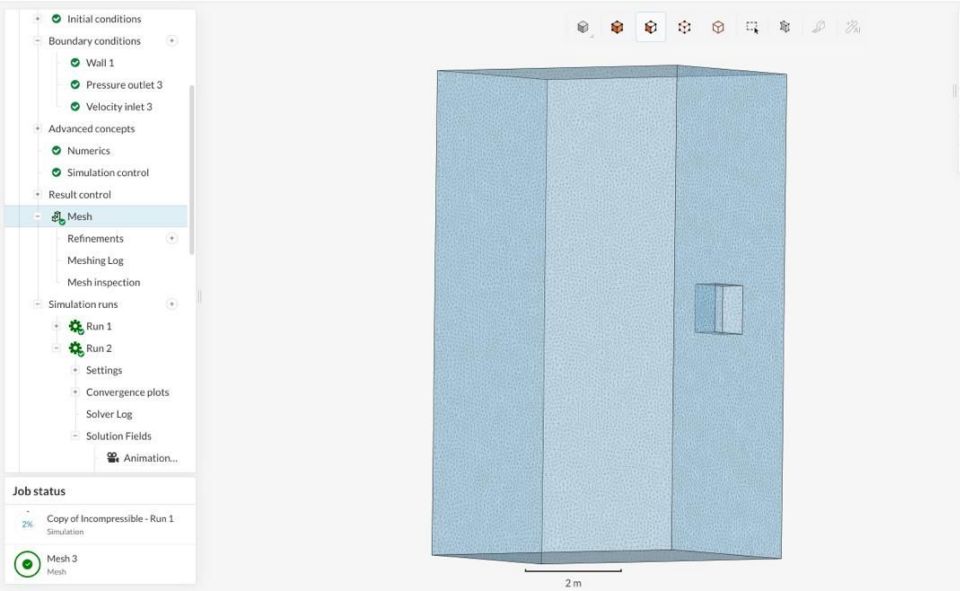
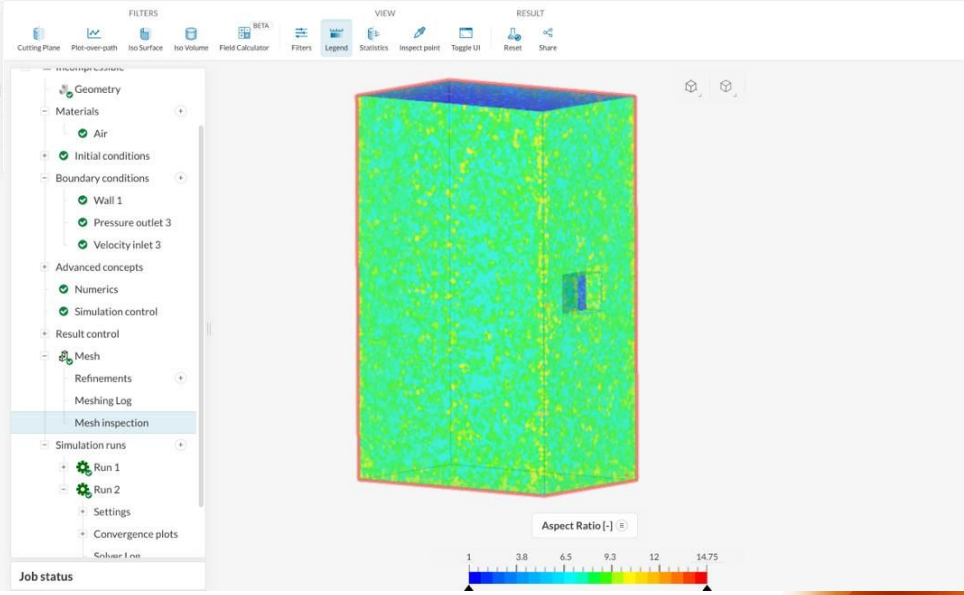


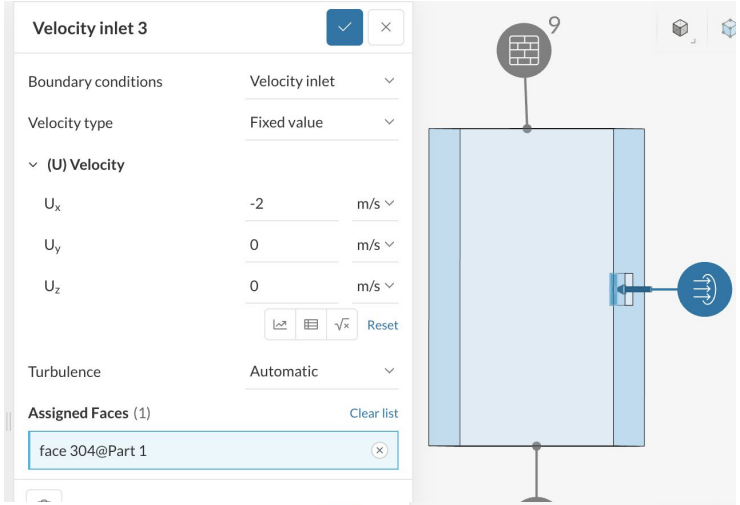
Figure 3. 4 Medium type of meshing of model Geometry

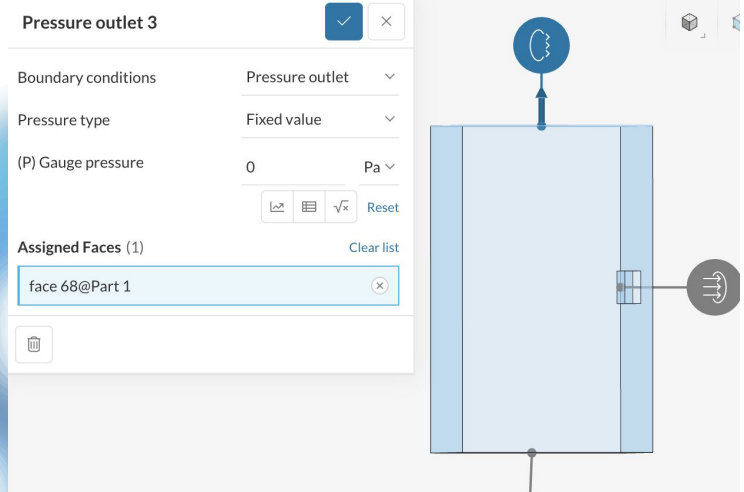
Generating Mesh

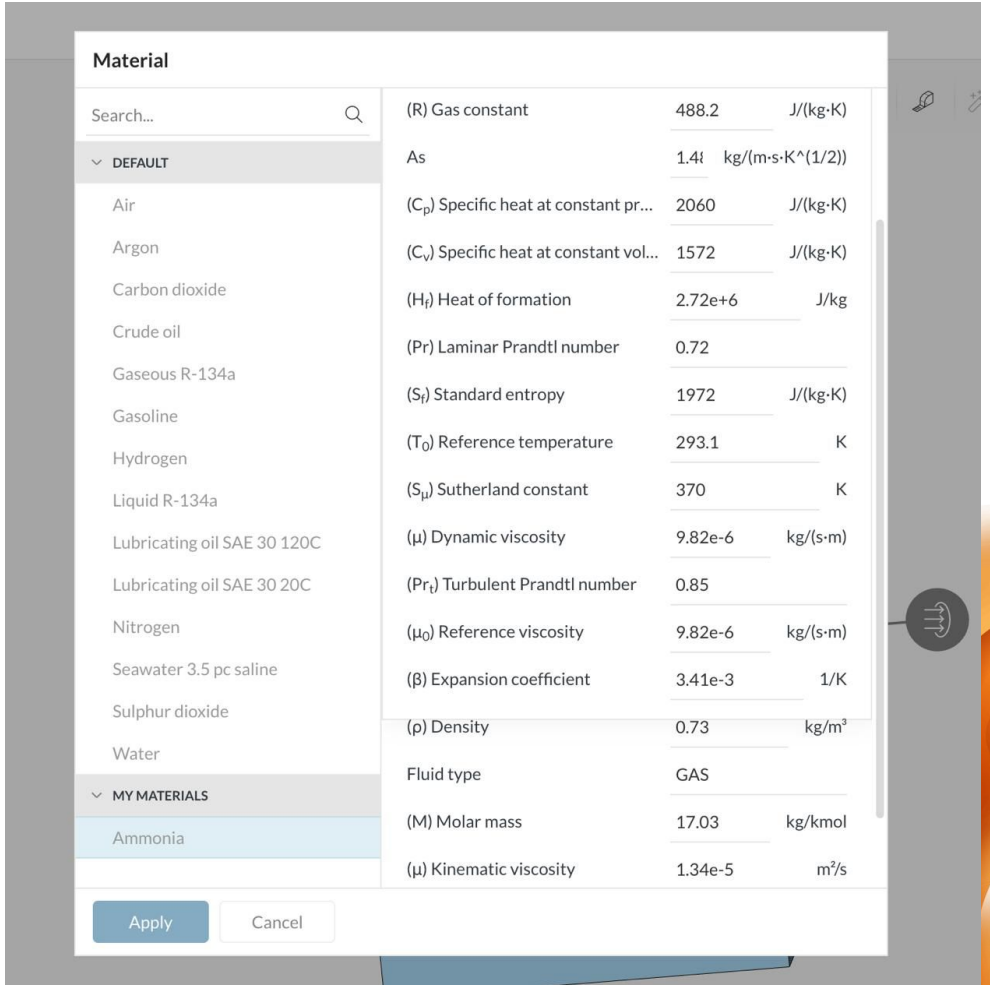


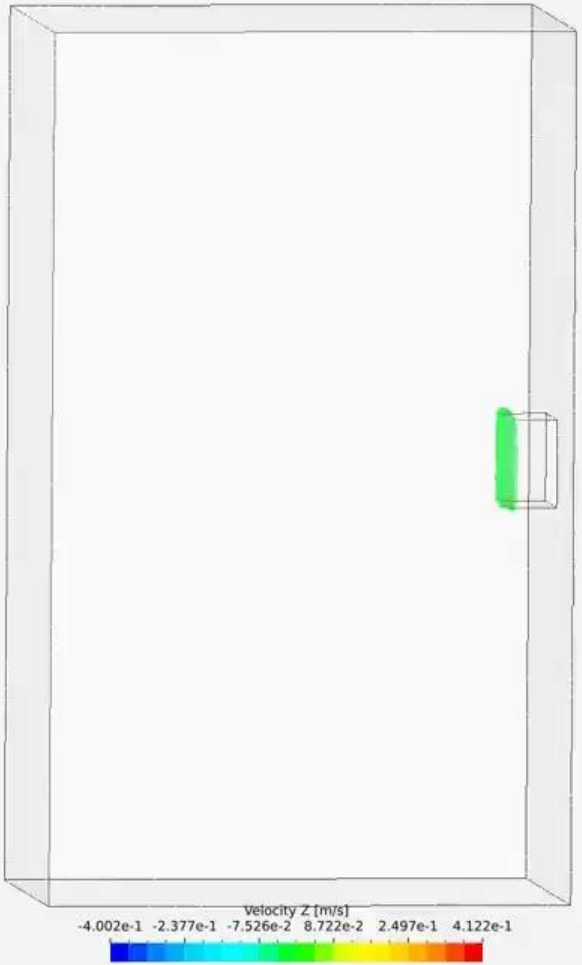


Optimization & Checks





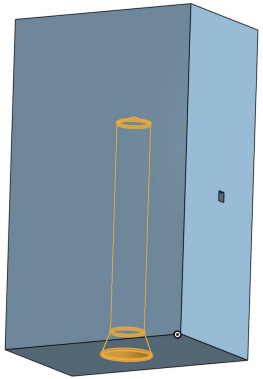




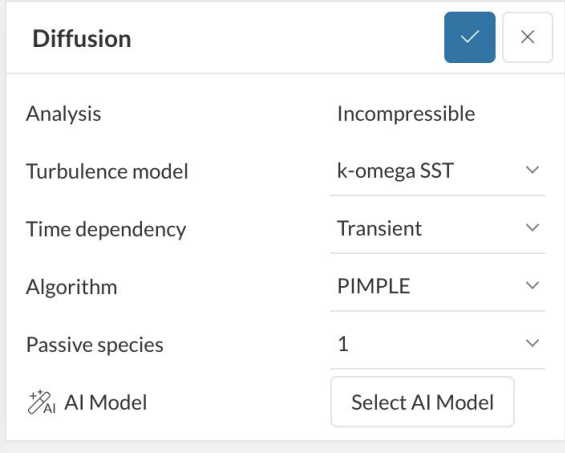
Run Time= 200s
Delta t= 1s
Control= Step

Simulation

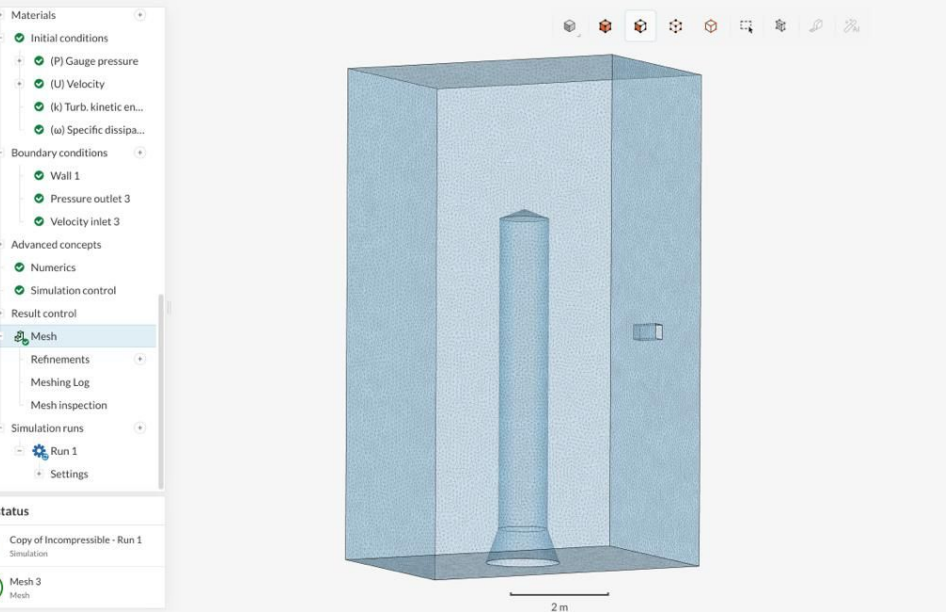
1.) Adding Obstacle (7x1m tower)

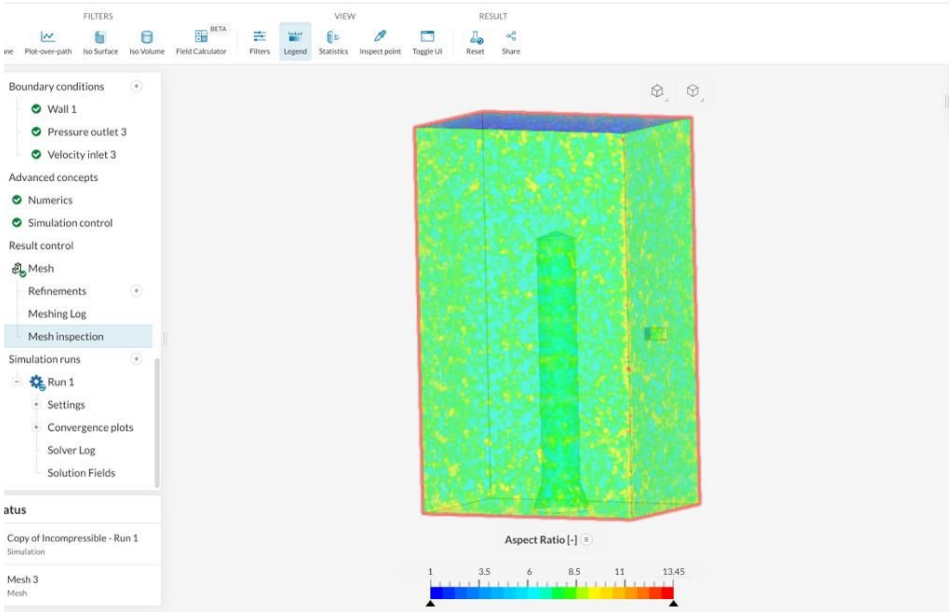


3.) Setting up simulation

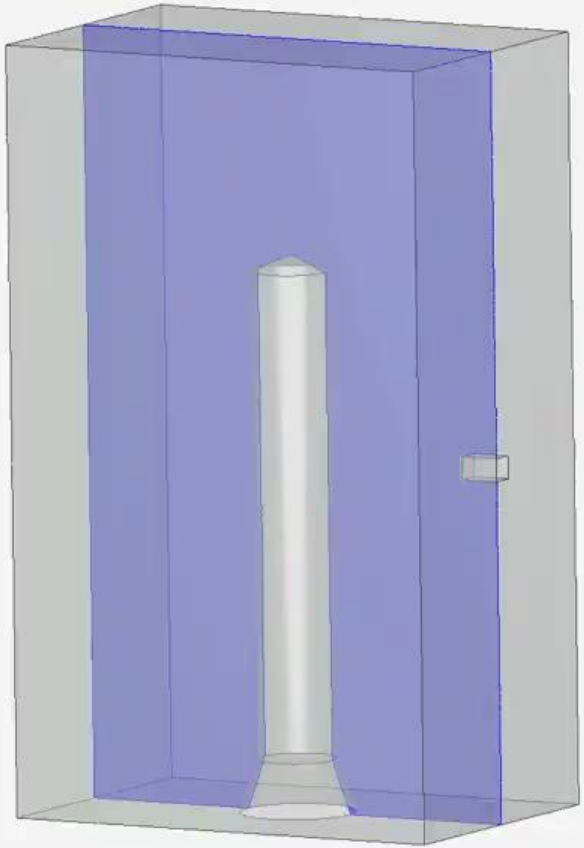


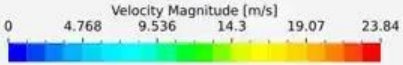
2.) Regenerating Mesh





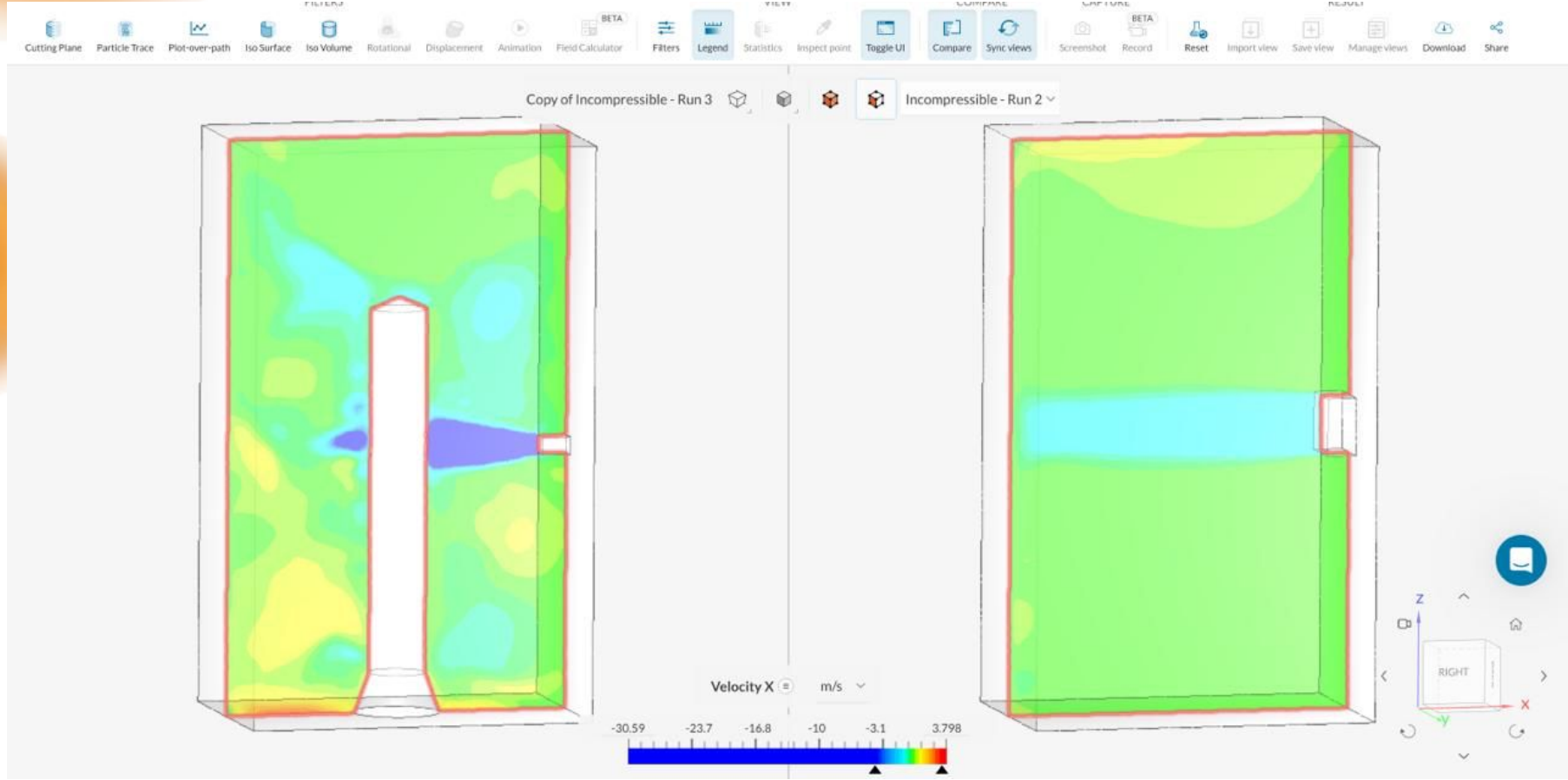
Velocity Flow over a cutting Plane



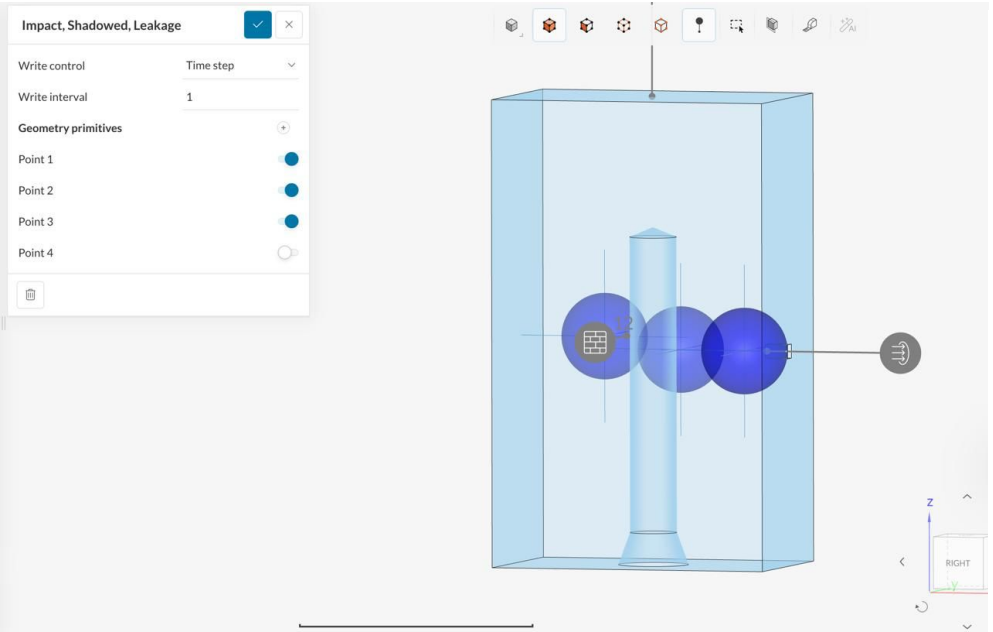


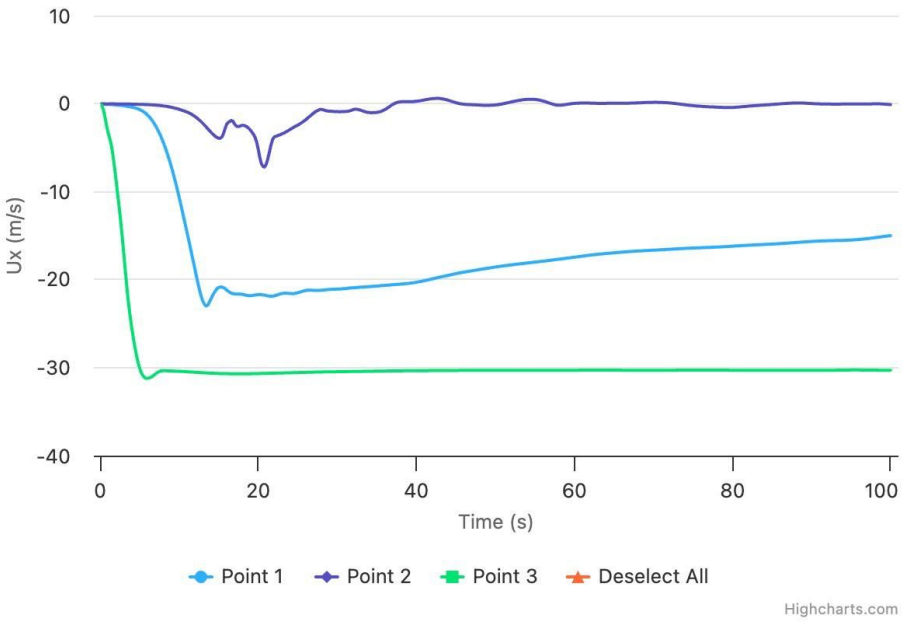
Additional Pressure inside the room along the flow of NH_3

Influence of obstacle



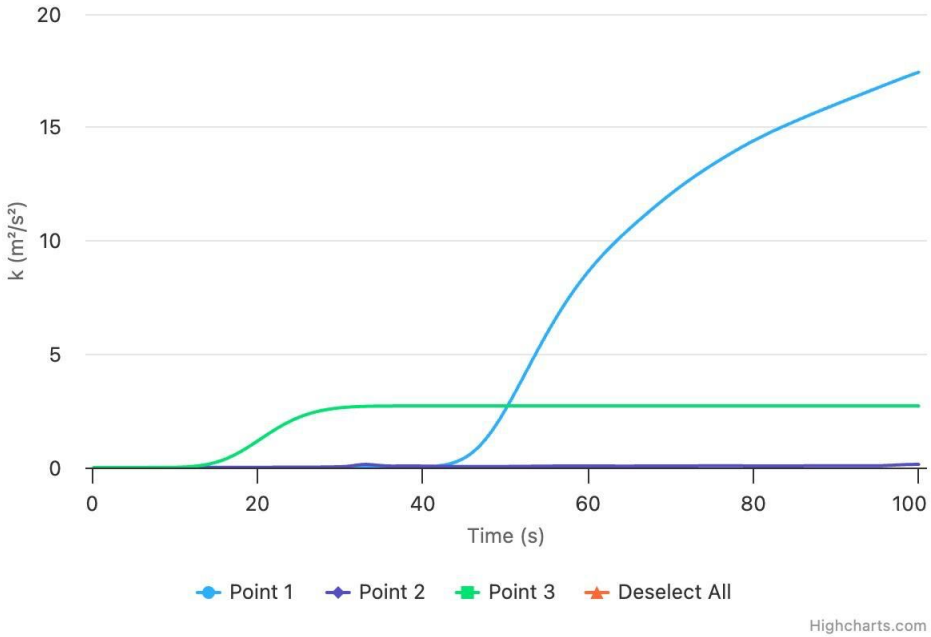
Plotting parameters across the volume

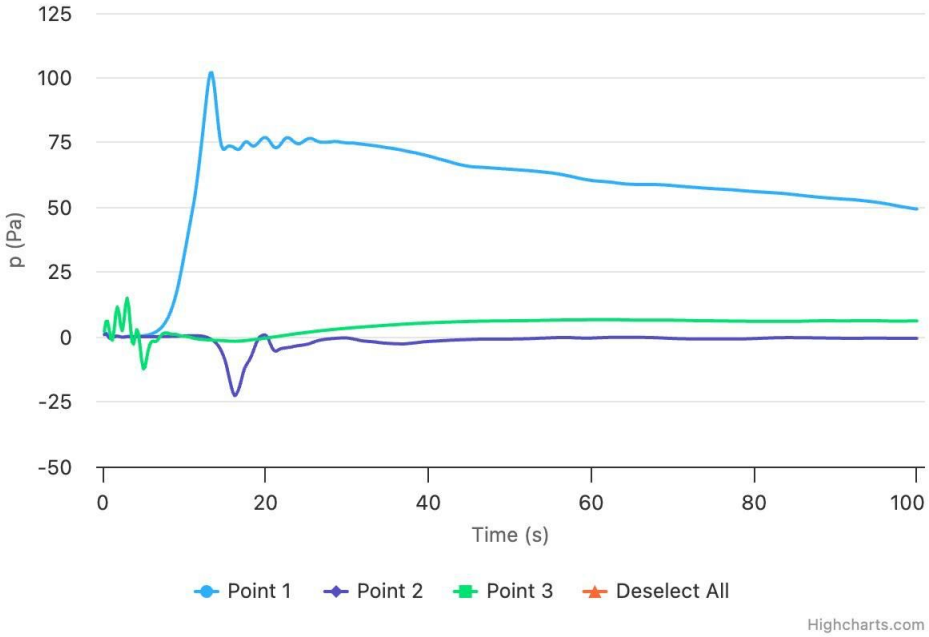




(Point 1= Impact, Point 2=Eclipsed, Point 3=Leakage)

Plotting parameters across the volume

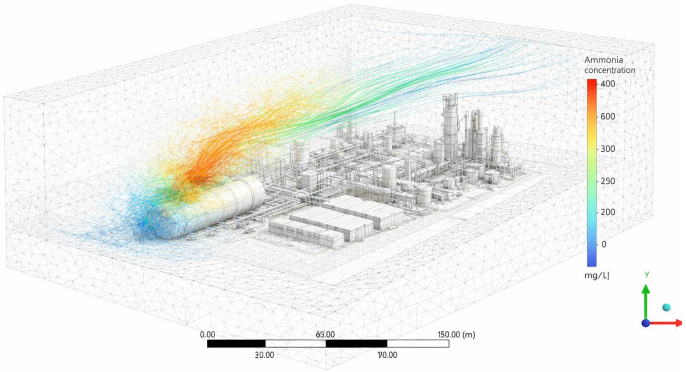




(Point 1= Impact, Point 2=Eclipsed, Point 3=Leakage)

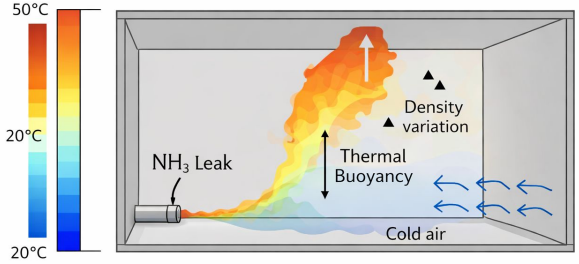
Further Improvements

1. **Use a realistic industrial geometry:** Instead of a simplified cuboidal room, the simulation could use a detailed CAD model of an industrial facility including machinery, pipelines, ventilation, ducts, storage equipment, etc. This would allow more realistic gas dispersion analysis.



2. **Include temperature effects:** Future work can incorporate temperature-dependent flow effects by switching from an incompressible model to a compressible or convective heat transfer model. This would allow the simulation to capture thermal buoyancy and density variations, providing a more realistic representation of ammonia dispersion in industrial environments.

Compressible model: Heat transfer affects density, causing upward flow due to thermal buoyancy



Temperature gradients drive ammonia plume upwards due to lower density at higher temperatures.

Thank You!